

MURALI SAI BUDDAKKAGARI VENKATA

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SUMMARY

AI Engineer building production agentic systems that reason, plan, and execute multi-step tasks: multi-agent orchestration in LangGraph, RAG pipelines (chunking, embeddings, retrieval, reranking, grounding), and tool integration via a custom MCP server. Applies prompt engineering, function calling, and rigorous evaluation to ship agents that are measured, not assumed to work. Shipped a multi-agent system validated by 106/113 tests against a live deployment, and a RAG pipeline evaluated across 9 metrics reaching 0.85+ faithfulness. Graduated with a Master of Science in Applied AI from Stevens Institute of Technology (May 2026).

TECHNICAL SKILLS

Programming Languages: Python, SQL

AI & Machine Learning: LLM Application Development, Agentic Systems & Tool Orchestration (MCP), Retrieval-Augmented Generation (Chunking, Reranking, Grounding), Prompt Engineering (Chain-of-Thought, Few-Shot), Function Calling, Multi-Agent Systems, Agent Evaluation (RAGAS), Guardrails & Reliability, Multimodal AI, NLP, Embeddings, Vector Databases, LangChain, LangGraph, PyTorch, TensorFlow, HuggingFace

Infrastructure & DevOps: FastAPI, ChromaDB, Docker, OpenTelemetry (Observability), Google Cloud Platform, Azure, PostgreSQL, GitHub Actions, CI/CD, Git, REST APIs, ReactJS, OpenAI API, Claude API, Gemini API

EXPERIENCE

Querkey Inc

Jun 2025 - Aug 2025

Data Science Intern

CA, US

- Developed and deployed a proof-of-concept real-time Patient Flow Tracking System using multimodal AI (Gemini 2.5 Flash) with chain-of-thought and few-shot prompting to improve routing efficiency, integrating a RAG pipeline that structured unstructured patient notes for real-time clinical insight retrieval.
- Cut production ML workflow failures 60% across 100K+ records by deploying a Dockerized observability stack (MLflow, FastAPI, ChromaDB vector store, HuggingFace monitoring), enabling real-time failure detection.

VIEW GROUP INDIA (FORMERLY VIEW SYNERGY)

Jan 2024 - Apr 2024

AI and Automation Intern

KA, IND

- As part of a cross-functional project team modernizing a live production platform, reviewed React frontend, .NET Azure Functions API, and MySQL stored-procedure code, tracing 2 recurring exceptions to root cause and surfacing SQL-injection vulnerabilities that fed into a 94-issue security assessment.
- Built and tested a new Roles/Designations CRUD feature end to end, including new MySQL stored procedures, a .NET API layer, and a React frontend, while documenting Microsoft Entra ID role setup for a proposed identity migration.

PROJECTS

SEC EDGAR RAG Filing Analyzer

- Deployed a production RAG system chunking and indexing 8,232 SEC 10-K filing chunks across 5 companies, verified by a 504-test regression suite run in CI and a 54-question eval suite (0.853 faithfulness, 0.936 citation accuracy, 0.981 refusal correctness), by building a FastAPI + ChromaDB embeddings pipeline with cross-encoder reranking, RBAC-enforced information barriers (FINRA Rule 2241 / SEC Rule 15g-1), and a serverless Google Cloud Run deployment (scale-to-zero).
- Built literal entity detection pinning retrieval to the named company's filing at the database level, per-entity retrieval budgets for comparative questions, and a bounded agentic retry with prompt-injection and MNPI/investment-advice output guardrails.

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Shipped a production multi-agent system (19 REST endpoints) that reasons, plans, and executes multi-step tasks end-to-end, validated by 106/113 passing tests against a live deployment, by architecting a 3-layer LangGraph hierarchy (planning supervisor, parallel specialists, reflective reviewer) with multi-model routing (GPT-4o, Claude Haiku) and two-tier Redis + Supabase pgvector memory.
- Built a custom MCP (Model Context Protocol) server integrating internal and third-party tools via APIs across 6 tools, instrumented end to end with OpenTelemetry distributed tracing, and a Human-in-the-Loop system spanning 4 escalation severity levels with full audit-trail logging.

PUBLICATIONS AND CERTIFICATIONS

- Adaptive Ensemble Detection for Medical Imaging, M.S. Capstone, Stevens Institute of Technology (Advised by Prof. Kevin Lu). A size-conditioned Weighted Box Fusion ensemble of Faster R-CNN, Mask R-CNN, and YOLO26 for cell-nucleus detection, reaching 0.71 mAP@0.5 on the 2018 Data Science Bowl benchmark.
- Best Research Paper Award, ICICC 2024 - WorkFit: Real-time Multimodal Posture Monitoring System using CNN-based Computer Vision and Flex Sensor Fusion.

EDUCATION

Stevens Institute of Technology

Master of Science, Applied Artificial Intelligence

May 2026

NJ, US

PES University

Bachelor of Technology, Computer Science (Specialization in ML)

May 2024

KA, IND